

Quiz 1 — CS0.101 Computer Programming — Monsoon '25 Semester

Questions compiled from given answer rubric

Q1. (5 marks) Express -56 in signed 8-bit binary (2's complement notation). Show all working steps.

Q2. (3 marks) What does the following code snippet output, and what does it compute?

```
printf("%d", ((x ^ y) < 0));
```

Q3. (3 marks) Which of the following for loop headers are valid in C? (Tick all that apply.)

1. `for (i = 0; i < 10 ; i++)`
2. `for ( ; i < 10 ; i++)`
3. `for (i = 0; ; i++)`
4. `for (i = 0; i < 10 ; )`
5. `for ( ; ; )`

Q4. (3 marks) What is the output of the following program?

```
int i = 1, j;
for ( ; ; ) {
    if (i)        j = --i;
    if (j < 10)    printf("CproQuiz", j++);
    else          break;
}
return 0;
```

Q5. (3 marks) What is the output of the following program?

```
int i, count;
count = 0;

for(i = 0; i < 2025; i++); {
    count++;
}

printf("%d", count);
```

Q6. (3 marks) The following code performs a cyclic left shift of an array `arr` of size `n` by `d` positions. Fill in the three blanks.

```
int p = 1;
while (_____) {                // Blank 1
    int last = arr[0];
    for (int i = 0; _____; i++) {    // Blank 2
        arr[i] = arr[i + 1];
    }
    _____                // Blank 3
    p++;
}
```

Q7. (3 marks) What is the output of the following program? (Note: `~` has higher precedence than `<<`.)

```
char x = 6;
char y = ~ x << 2;

printf("%d\n", y);
```

Q8. (3 marks) What is the output of the following program?

```
signed char a = 100, b = 30;
signed char c = a + b;

printf("a+b = %d\n", c);
```

Q9. (3 marks) What is the complete output of the following program?

```
int x = 0, y = 5;

if (x++ && ++y)
    printf("Branch 1\n");

printf("x = %d, y = %d\n", x, y);

if (++x || y++)
    printf("Branch 2\n");

printf("x = %d, y = %d\n", x, y);
```

Q10. (3 marks) What is the output of the following program?

```
int a = 3, b = 7, c = 2;
if (a++ > 3) {
    if (--b < 7) {
        printf("P5: X\n");
    } else {
        printf("P5: Y\n");
    }
} else {
    if (c++ == 2) {
        printf("P5: Z\n");
    } else {
        printf("P5: W\n");
    }
}
```

Q11. (3 marks) What is the output of the following program?

```
int i, j;
for (i = 1; i <= 3; i++) {
    for (j = i; j <= 3; j++) {
        printf("%d ", i + j);
    }
    printf("\n");
}
```

Q12. (3 marks) What is the output of the following program?

```
int a[] = {4, 3, 5, 2, 1};
int i = a[0];
while (i != 1) {
    printf("%d ", i);
    i = a[i - 1];
}
```